INCH-POUND
MIL-M-38510/330B
AMENDMENT 3
August 14, 1997
SUPERSEDING
AMENDMENT 2
27 October 1987

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL,
NAND GATES, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION.

This amendment forms a part of MIL-M-38510/330B, dated 5 April 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title, delete and substitute as printed above.

1.2.3, delete case outlines X and Y and respective descriptions, and substitute:

"2 C-2 (20-terminal, .350" x .350"), square chip carrier package
X C-2A (20-terminal, .350" x .350"), square chip carrier package"

1.3, Thermal resistance, junction-to-case (Θ_{JC}): Delete and substitute as follows:

"Cases A, B, C, D, X and 2 ---------- (See MIL-M-38510, appendix C)

1.3, Junction temperature (T_J),: Add " 2/" after (T_J).

After footnote 1/ add the following footnotes:

"2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883."

PAGE 2

1.4, after supply voltage,: add "(V_{CC})".

1.4, delete "Normalized fanout (each output) 2/:" and substitute "Normalized fanout (each output) 4/:".

3.2.3, delete and substitute:

* "3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity (DSCC-VA) prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity (DSCC-VQ) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the preparing activity and will be available upon request."

Bottom of page, delete footnote "2/" and substitute footnote "4/"

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PAGE 3

TABLE I, I_{IL} min limits column, delete “-.03” and substitute “0”.

* TABLE I, For device 02 only,
I_{OD}, min = 60 mA ---------- Min = 50 mA.

PAGE 4

TABLE II, group A test requirements, class B devices, add “10, 11”.

TABLE II, Interim electrical parameters, delete “(pre burn-in)”.

TABLE II, delete “Additional electrical requirements for group C periodic inspections” in its entirety.

3.6, Marking, delete second sentence.

4.2a(1): Delete “Test condition D or E” and substitute “Test condition D, E, or F”.

PAGE 5

4.2c, delete and substitute:

“c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510”.

4.4.3b, delete in its entirety.

4.4.3c(1): Delete “D or E” and substitute “D, E, or F”.

4.4.3c(3): Delete “appendix B of MIL-M-38510.” And substitute “method 1005 of MIL-STD-883.”

PAGES 7, 8, 9, AND 10

FIGURE 1: Delete “Case X, Y” and substitute “Case X, 2”.

PAGES 12, 13, 14, AND 15

FIGURE 3, line 2, delete “and Y” and substitute “and 2”.

FIGURE 3, line 3, delete “for cases X and Y” and substitute “for cases X and 2”.

PAGE 15

* FIGURE 3: Add optional burn-in circuit as printed on page 5 of this amendment.

PAGES 17, 18, 19, 20, 21, 22, 23, AND 24

TABLE III, column four: Delete “Cases X, Y” and substitute “Cases X and 2”.

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PAGES 18, 20, 22, AND 24

TABLE III, Subgroup 9, t_{PHL}, (device type 01, test nos. 55-62), (device type 02, test nos. 51-56), (device type 03, test nos. 57-65), (device type 04, test nos. 49-56), minimum limits column: Delete "2.0" and substitute "1.5".

PAGE 18

TABLE III, device type 01: Delete note 2/ and substitute:
"2/ I_{IL} limits shall be as follows:

Test	Min/max limits in mA			
	Circuits			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.06	-.03/-.60	0/-0.3

"

PAGE 20

Table III, device type 02: Delete note 2/ and substitute:
"2/ I_{IL} limits shall be as follows:

Test	Min/max limits in mA			
	Circuits			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.06	-.03/-.60	0/-0.3

"

PAGE 24

Table III, device type 04: Delete note 2/ and substitute:
"2/ I_{IL} limits shall be as follows:

Test	Min/max limits in mA			
	Circuits			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.06	-.03/-.60	0/-0.3

"

PAGE 25

6.1, delete in its entirety.

6.4, delete "Electrical ground (common terminal)" and substitute "Ground zero voltage potential".

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Table IV, delete and substitute:

"TABLE IV. Manufacturers' designations.

Device type	Manufacturer			
	Circuit A	Circuit B	Circuit C	Circuit D
	Fairchild Semiconductor	Motorola Incorporated	Signetics Corporation	Texas Instruments
01	X	X	X	X
02	X	X	X	X
03	X	X	X	
04	X	X	X	X

"

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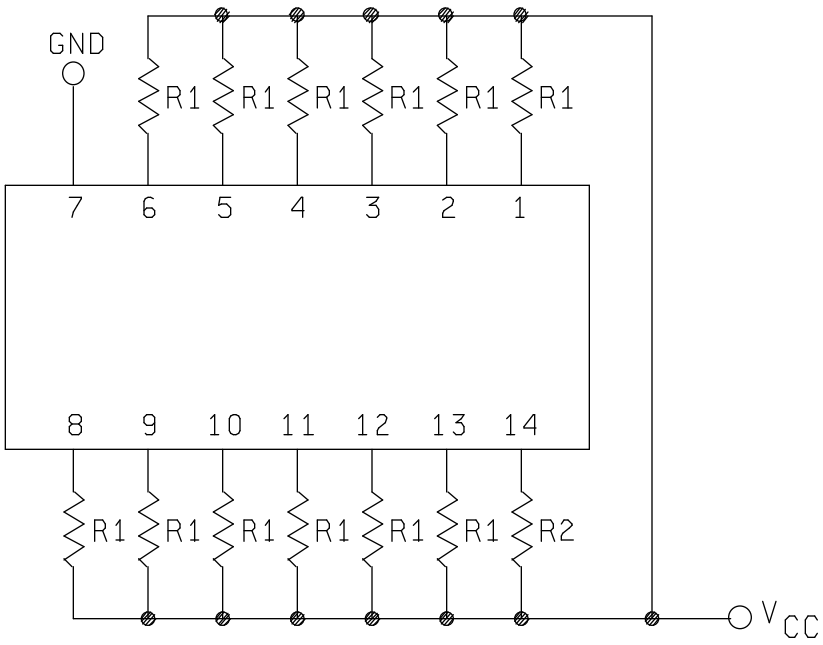
Device types 01, 02, 03, and 04
Excluding case 2

High-temperature (175°C) burn-in (optional circuit)



NOTES:

1. High temperature (175°C) condition F burn-in circuit only.
2. R1 = 2k Ω, R2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 3. Burn-in and life test circuit - Continued.

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